

# OpenSMART: Single-cycle Multi-hop NoC Generator in BSV and Chisel

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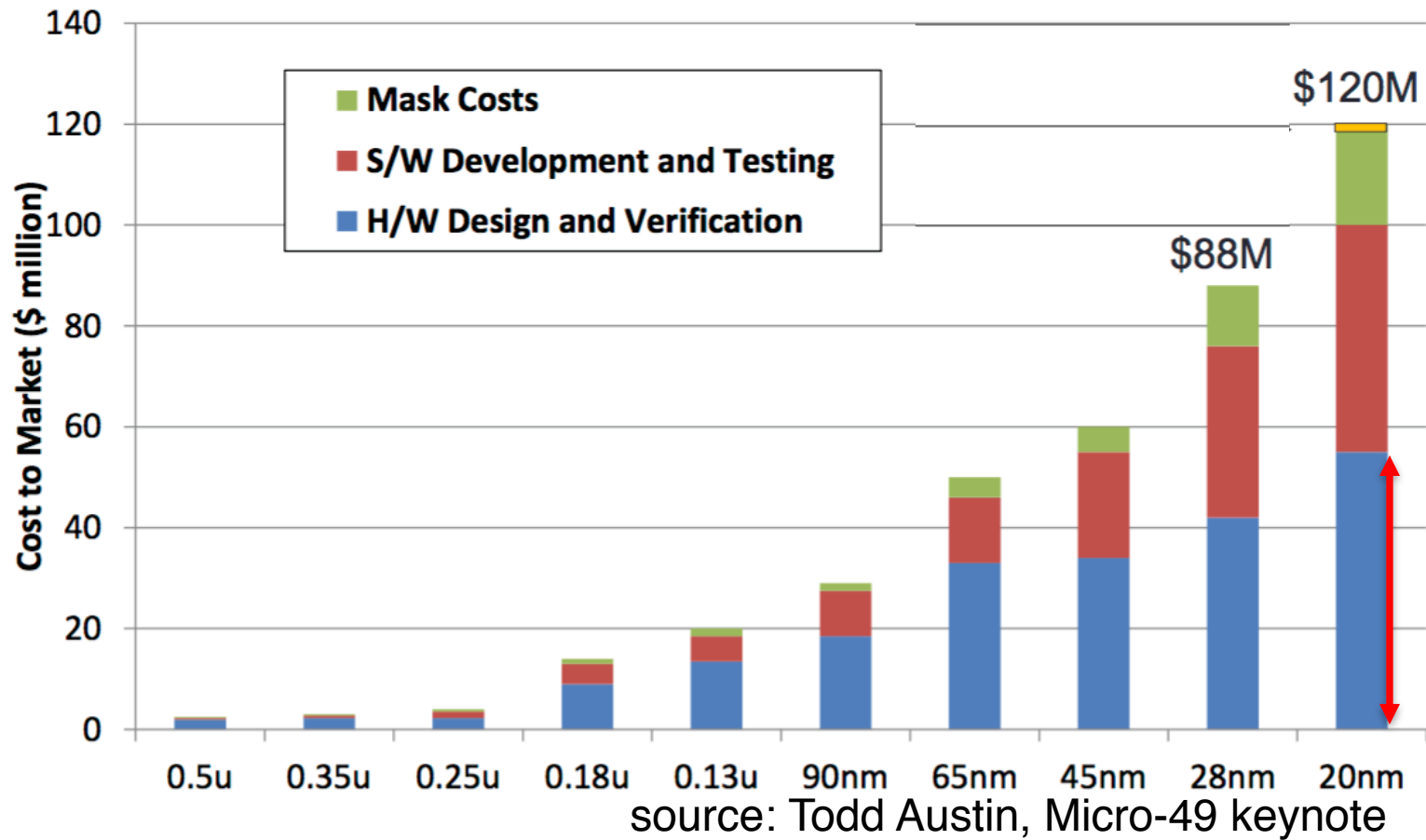
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Georgia Institute of Technology  
Synergy Lab (<http://synergy.ece.gatech.edu>)

**hyoukjun@gatech.edu**

April 25, 2017

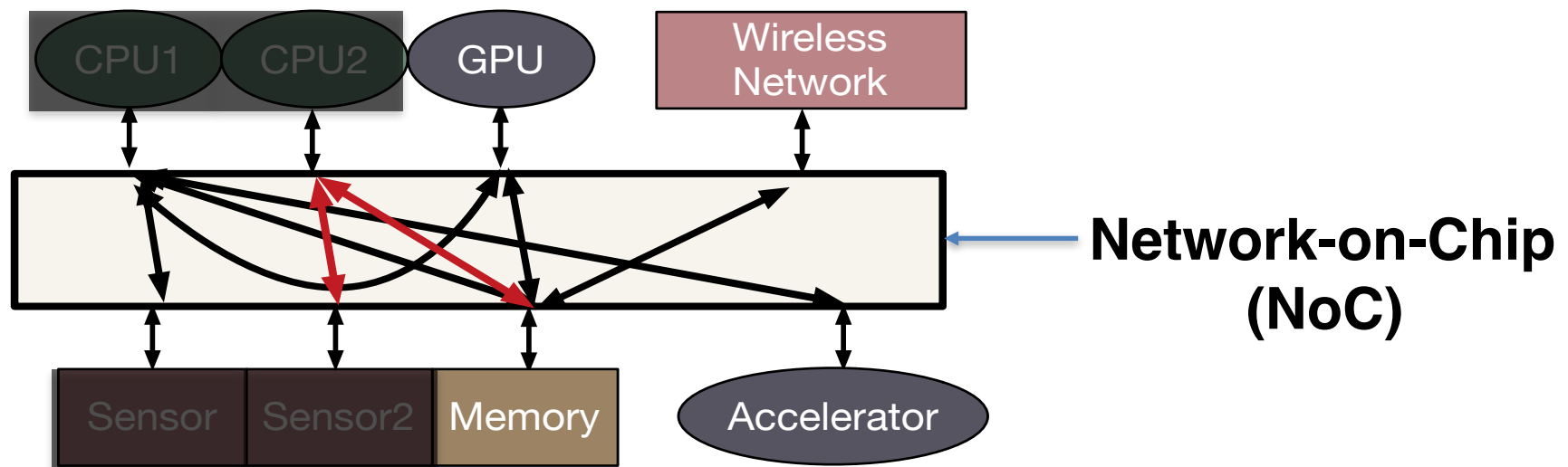
# Hardware Development Cost



- Low cost challenge

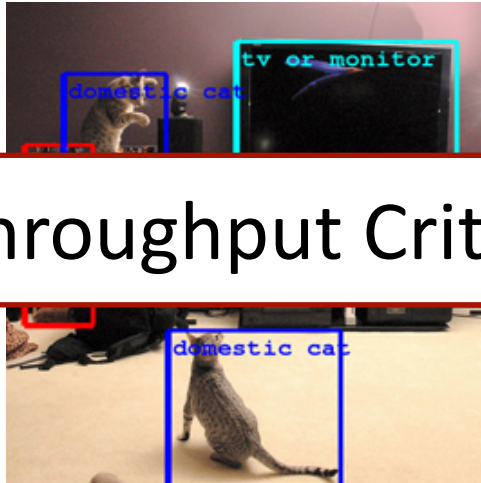
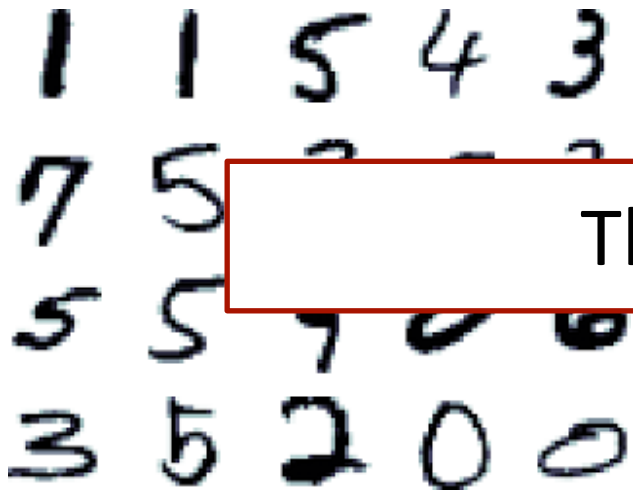
# Many-IP Heterogeneous System

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- **Scalability challenge**
- **Flexibility challenge**

# Diverse System Requirements



Throughput Critical



Latency Critical

source: MNIST, Engadget, TheStack

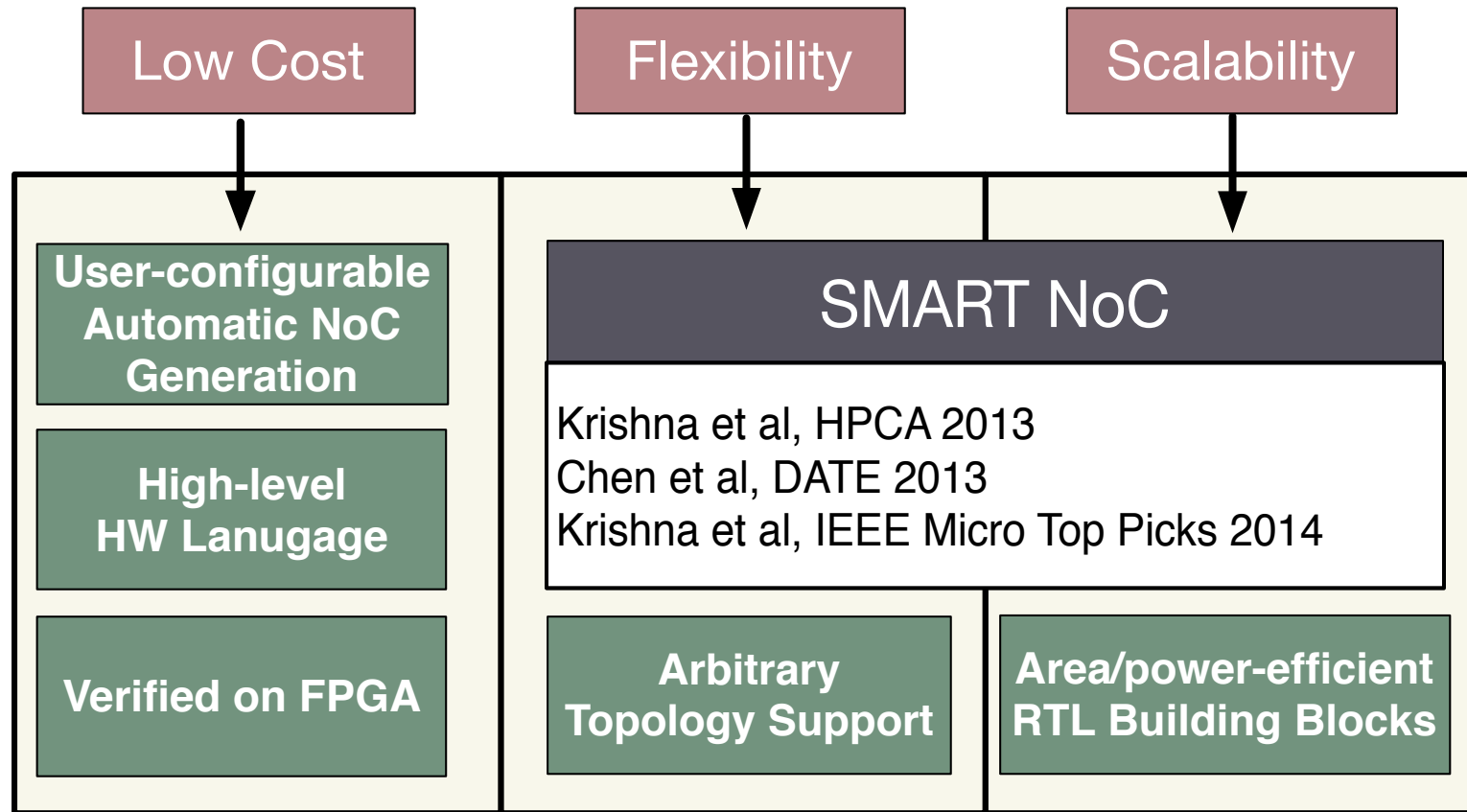
# Challenges for NoCs

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- **Low-cost**
  - Low design/verification costs of custom/generic NoCs
  - Design Automation of high-performance, low-energy NoCs
- **Scalability**
  - Many-IP heterogeneous system support
  - Low latency
  - Low energy
  - Low area
- **Flexibility**
  - Diverse connectivity
  - Diverse latency/throughput requirements

# OpenSMART

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## OpenSMART

# Outline

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- **Motivation: Scalable, Flexible, and Low-cost NoCs**
- **Background: SMART NoCs**
- **OpenSMART**
  - Design Flow
  - Building Blocks
  - Walk-through Examples
- **Case Studies**
  - Mesh vs. SMART
  - High-radix vs. Low-radix
- **Conclusions**

# Outline

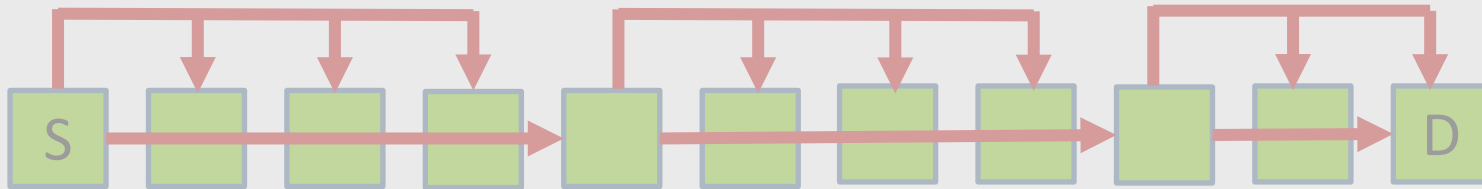
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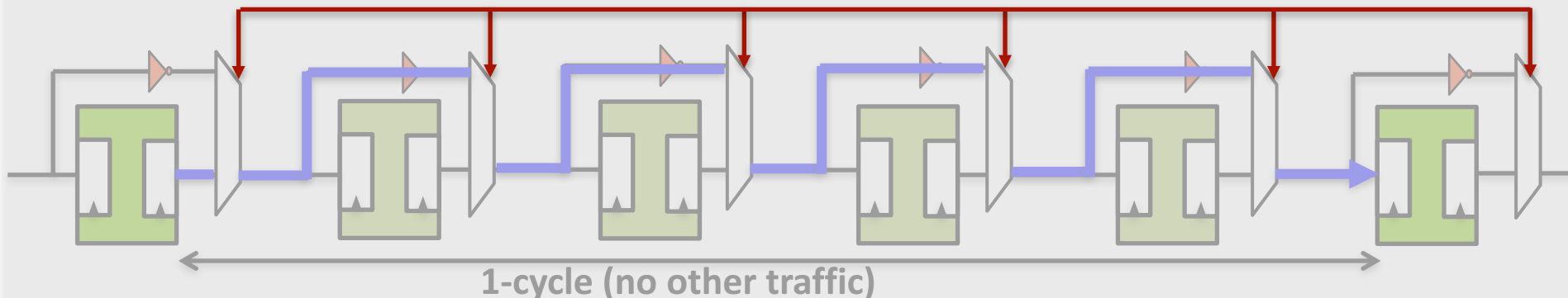
# SMART NoC

- **S**ingle-cycle **M**ulti-hop **A**synchronous **R**epeated **T**raversal

SSR (SMART Setup Request) SSR (SMART Setup Request) SSR (SMART Setup Request)



**SMART:** achieve the performance of *dedicated* connections over a network of *shared* links



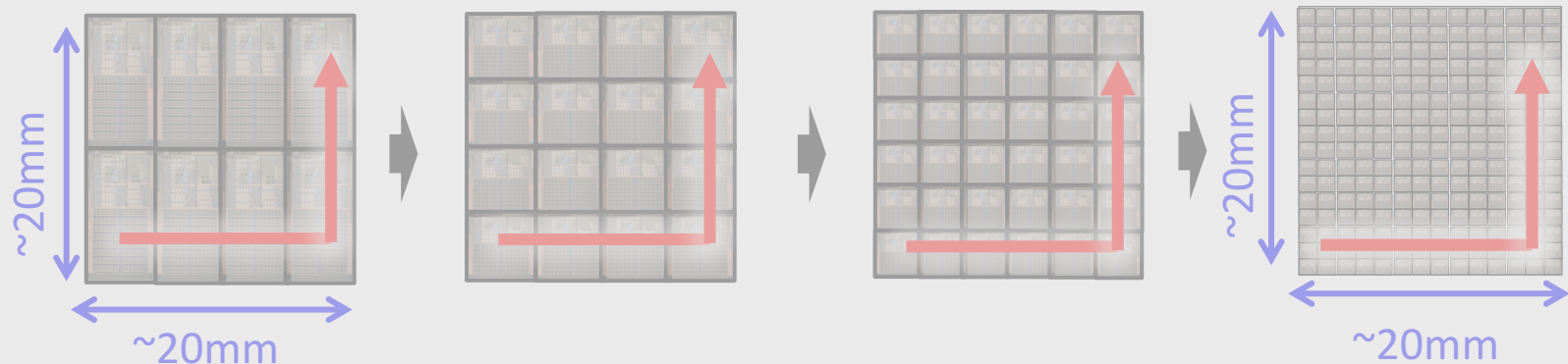
# Is 1-cycle Network Possible?

Is wire fast enough to support 1-cycle network?

Yes

- Wire traversal length within 1ns (1Ghz): **10-16mm** 😊
- Wire delay over technology: **constant** 😞
- Chip dimension: **remain similar (~20mm)** 😊

On-chip wires are fast enough to transmit across the chip **within 1-2 cycles** at 1GHz even if technology scales



# Features of SMART

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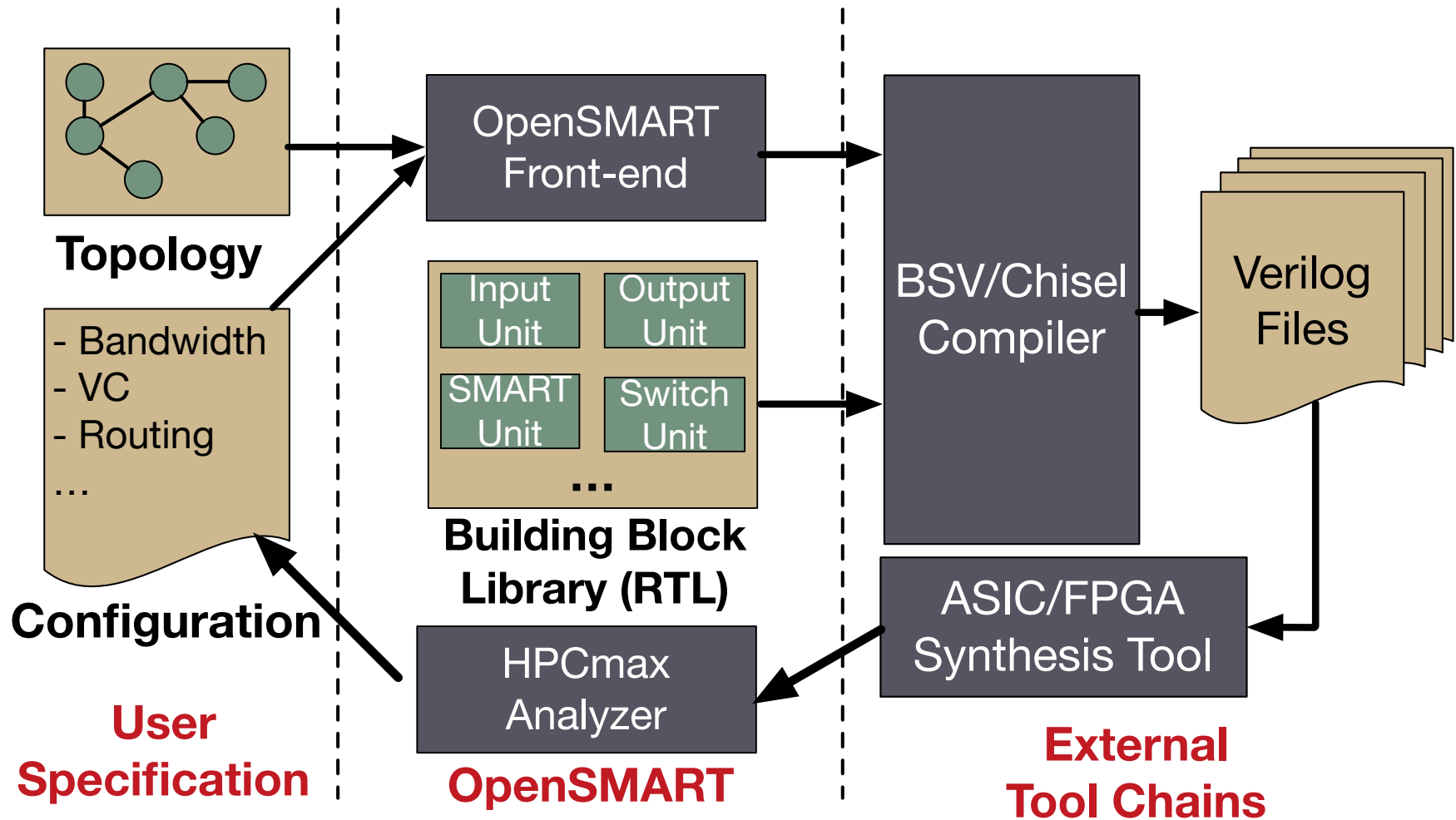
- **Low latency network**
  - **Dynamic** bypass of intermediate routers **between any two routers**
  - Limit: **HPCmax** (hops per cycle max), *maximum number of “hops” that the underlying wire allows the flit to traverse within a clock cycle*
- **Separate control path**
  - HPCmax bits from every router along each direction
  - Arbitration of multiple bypass requests on the same link
  - **No ACK** required

# Outline

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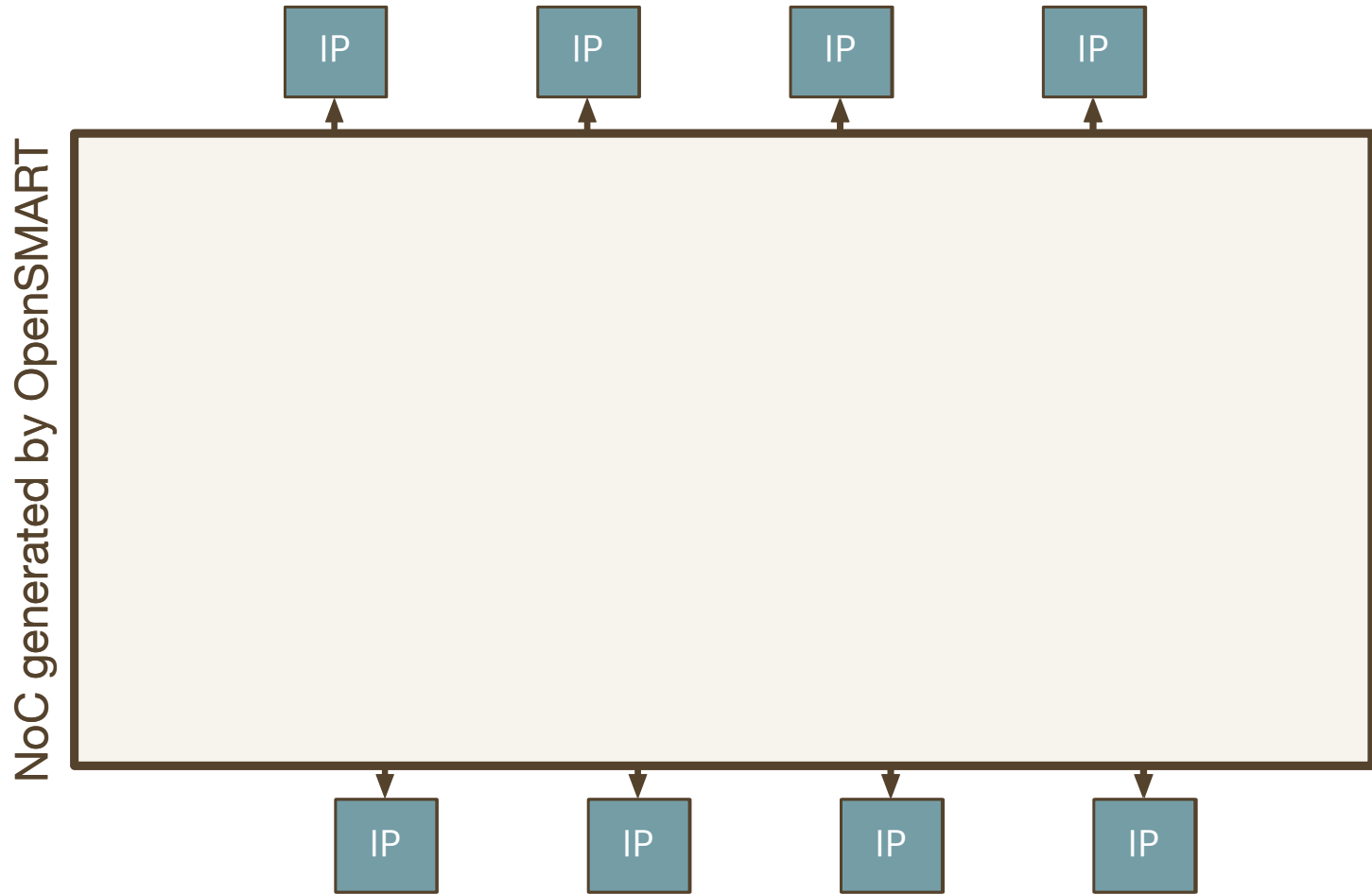
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# OpenSMART Design Flow



# OpenSMART NoC

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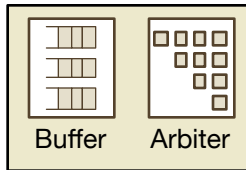
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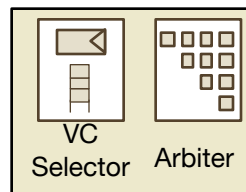
# OpenSMART Building Blocks

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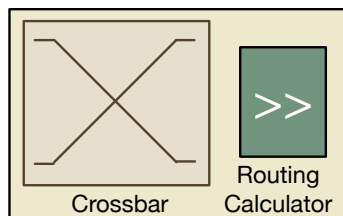
Input Unit

input Buffer + input VC arbitration



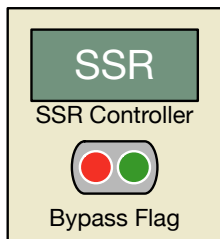
Output Unit

output VC selection + output port arbitration  
+ credit management



Switch Unit

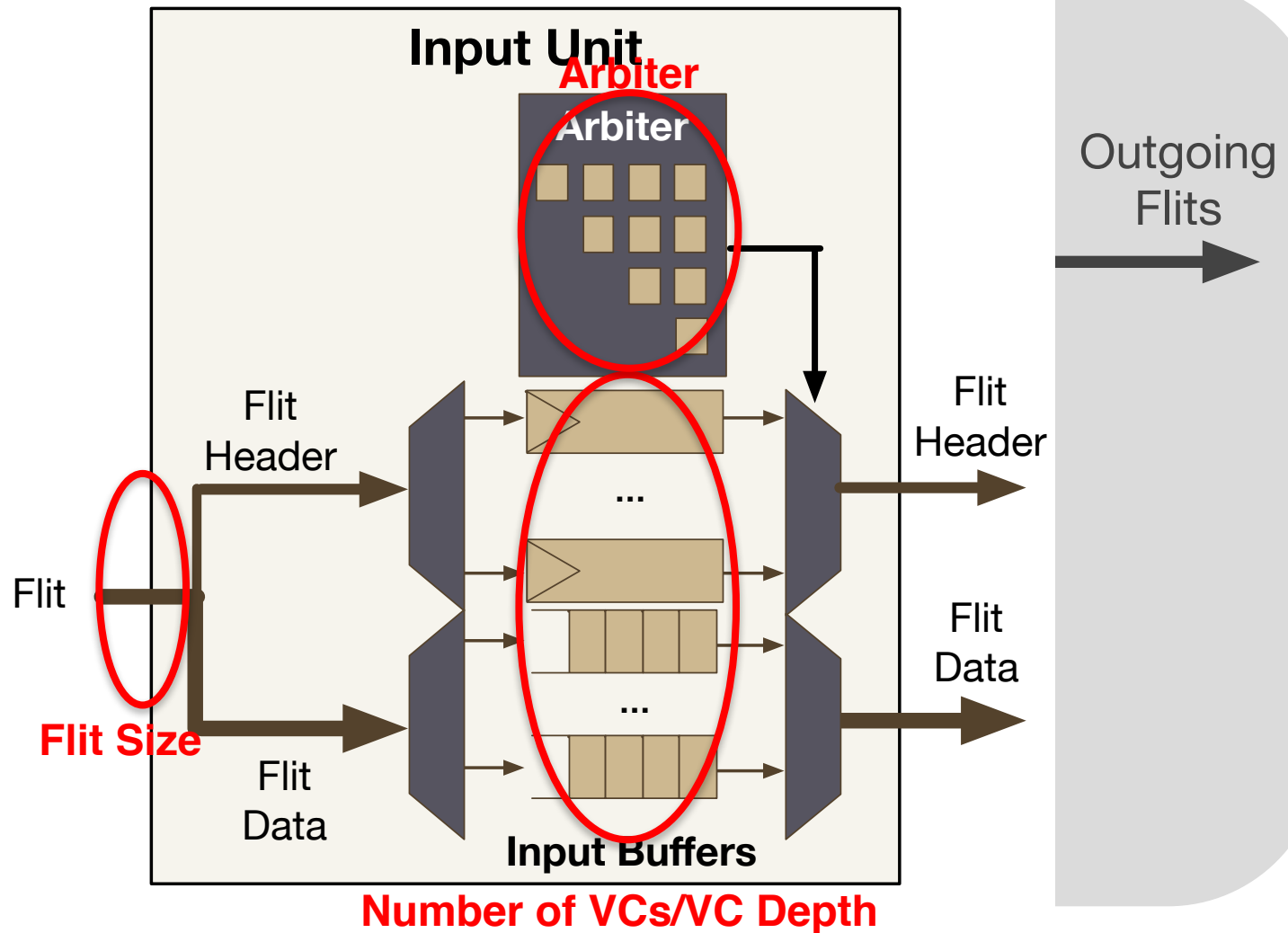
switching (via crossbar) + routing calculation



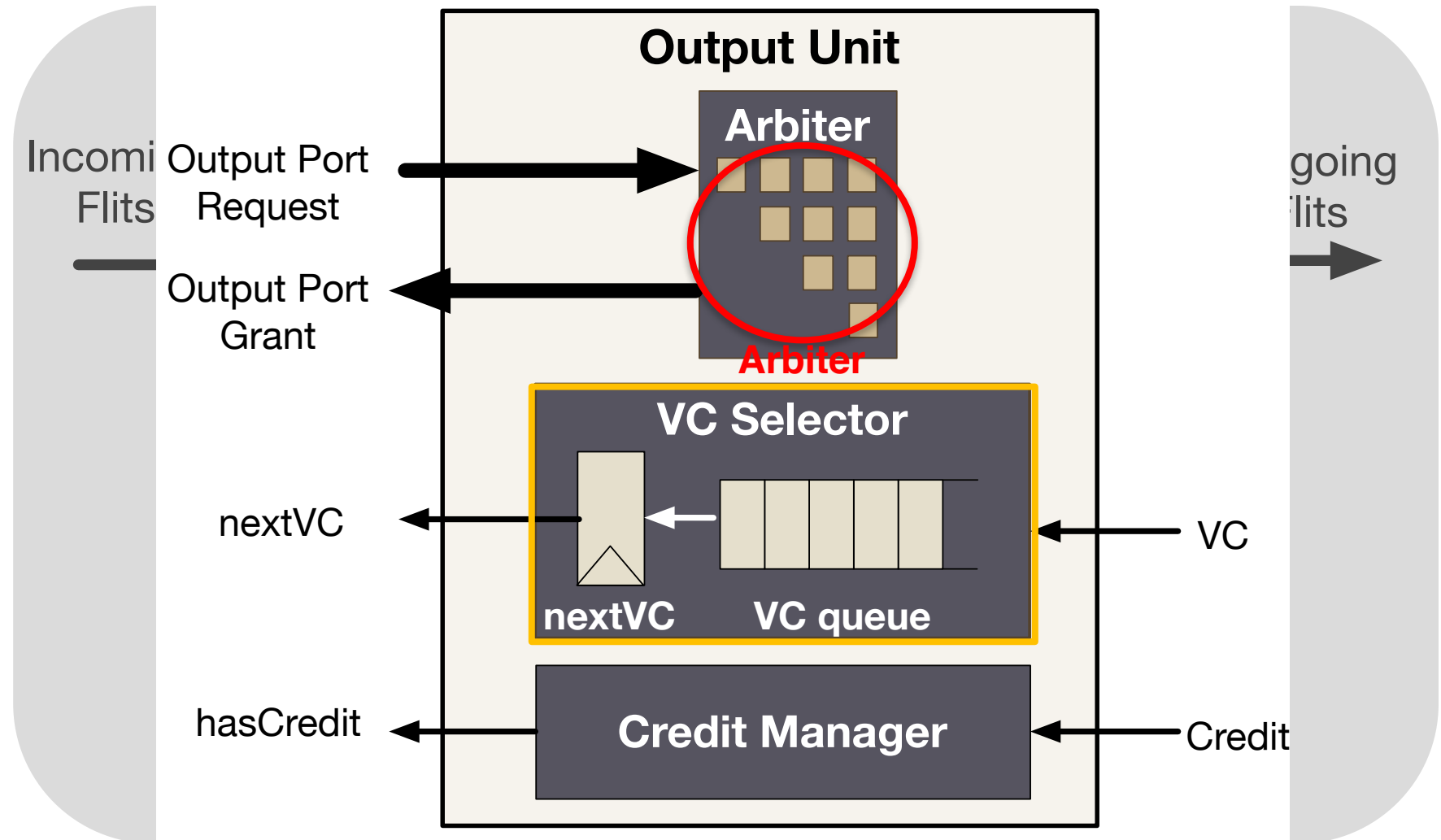
SMART Unit

SSR communication & arbitration + bypass flag

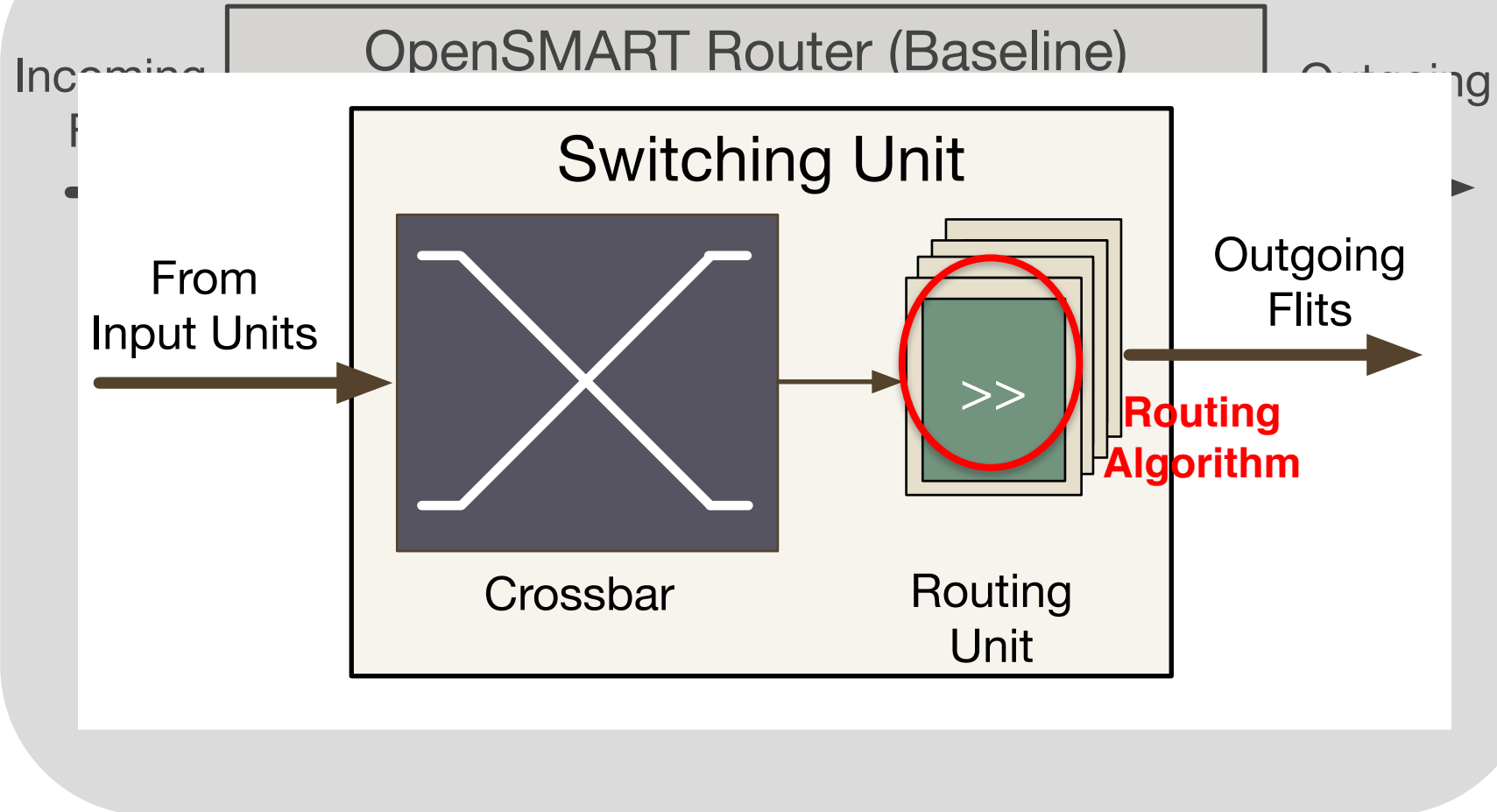
# OpenSMART Router



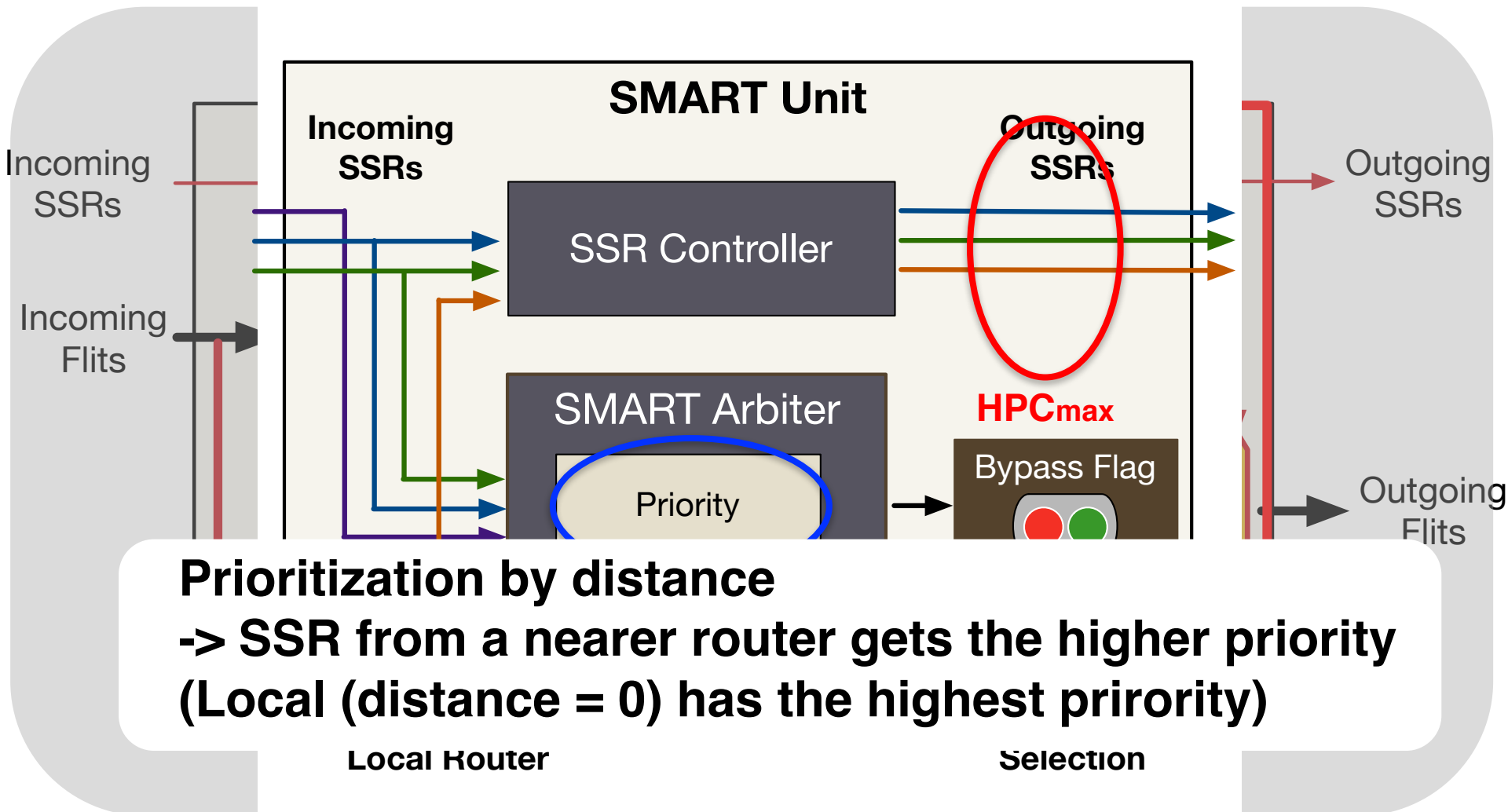
# OpenSMART Router



# OpenSMART Router

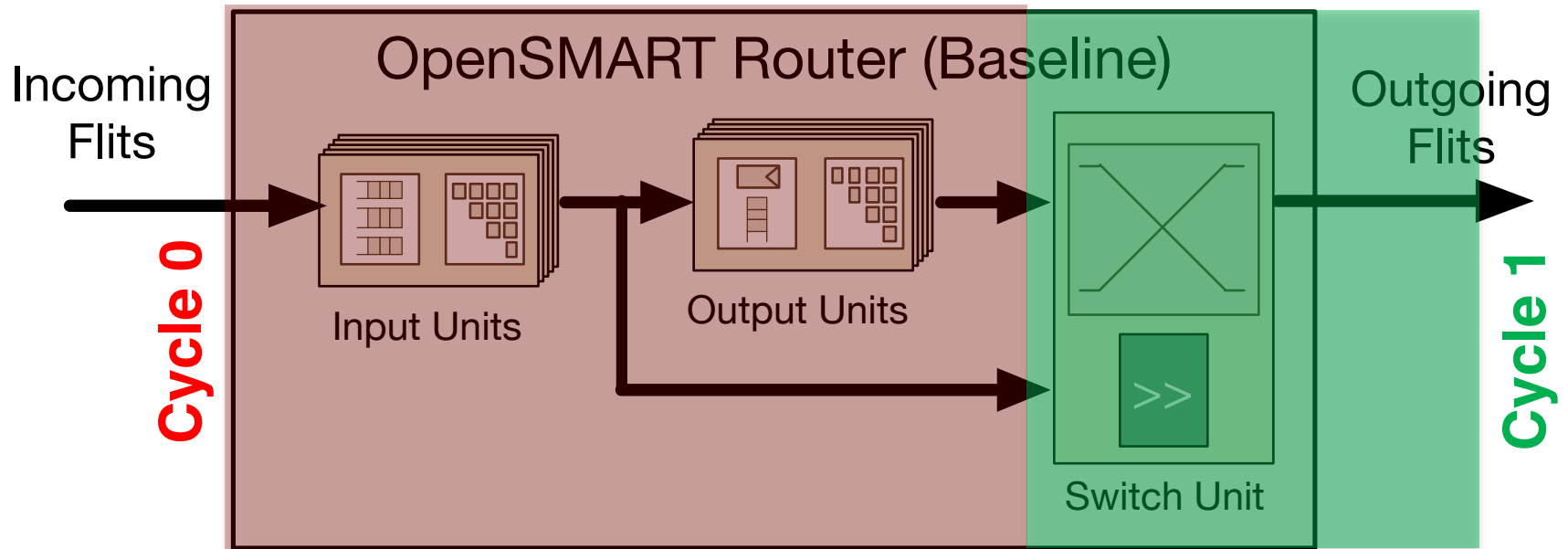


# OpenSMART Router (SMART)

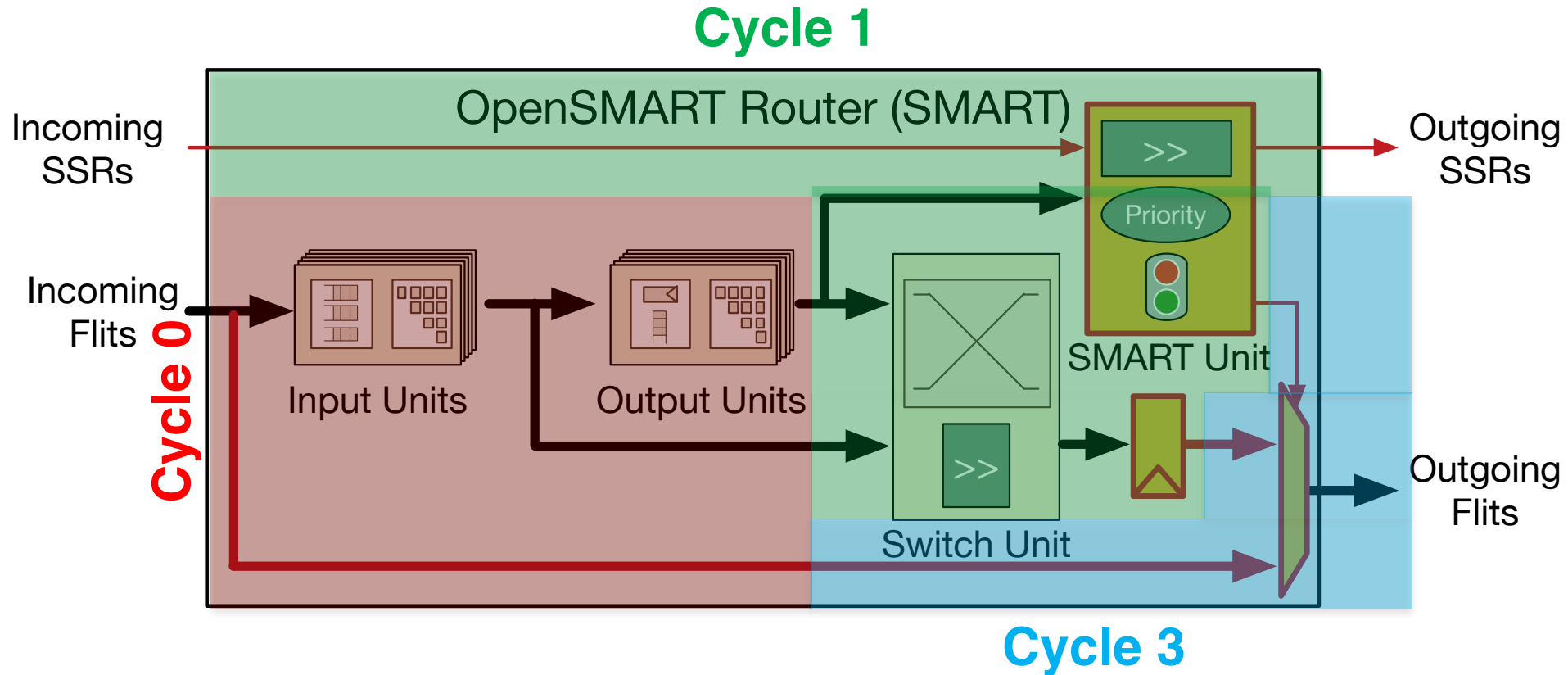


# OpenSMART Router (1cycle)

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# OpenSMART Router (2cycle/SMART)



# Outline

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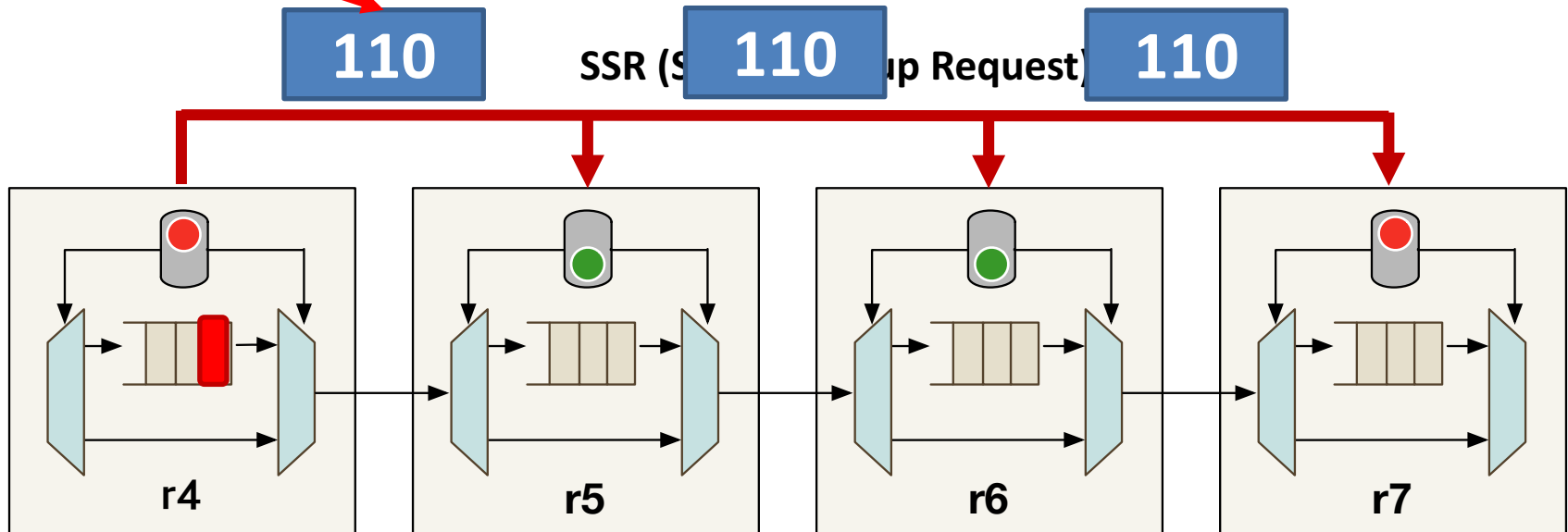
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# Walk-through Example 1

- Router r4 sends a flit to router r7
- $HPC_{max} = 3$

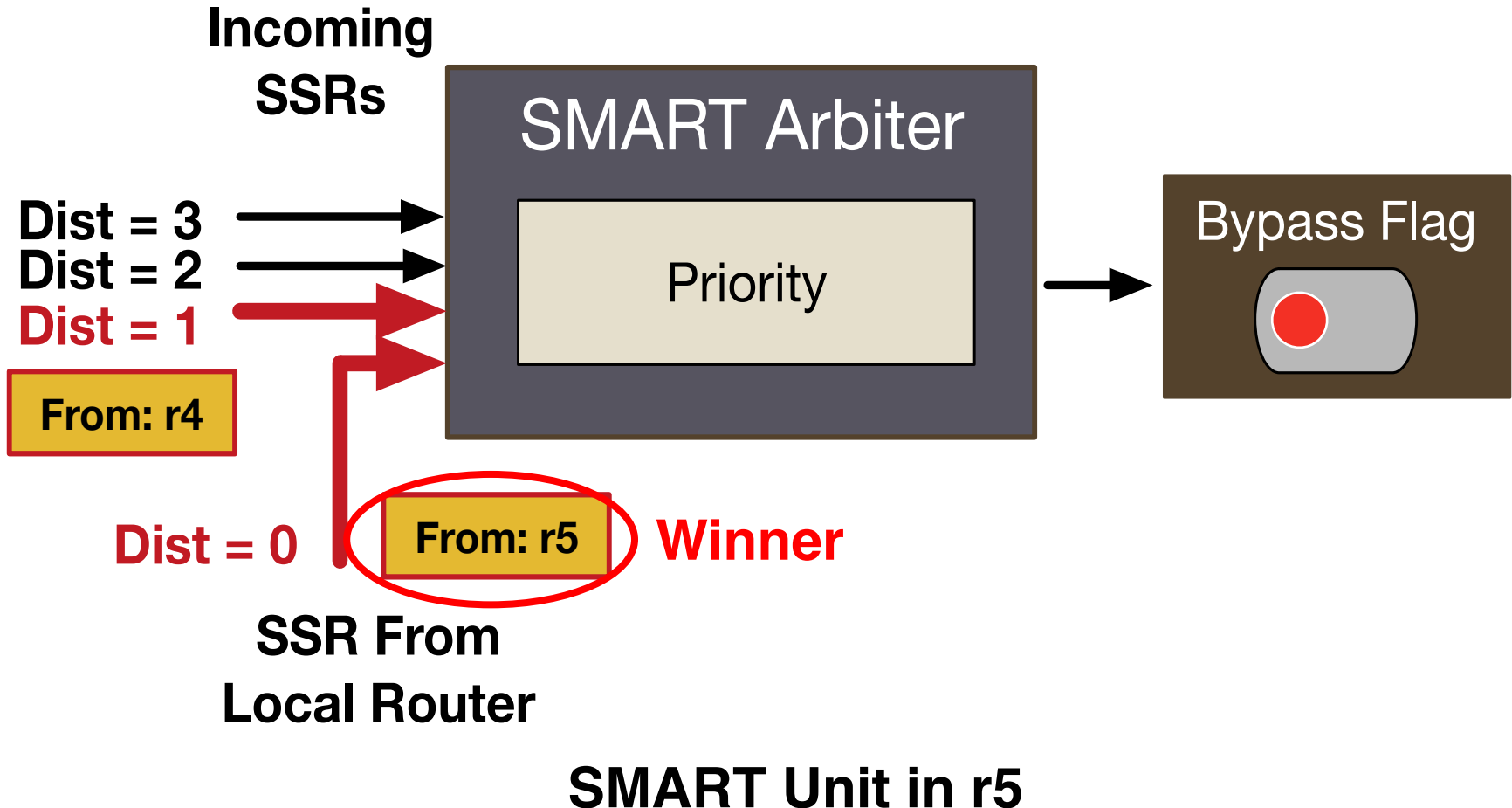
bypass, bypass, stop

Cycle Mult SSR Bypass



# Walk-through Example 2

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# OpenSMART: Features

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- **Language**
  - **BSV** and **Chisel**
- **Flow control**
  - **VC** and **SMART**
- **Buffer Management**
  - **Credit-based** buffer management
- **Router Microarchitecture**
  - **1- and 2-cycle** state-of-the-art packet switching router
  - **SMART** router

# OpenSMART: Features

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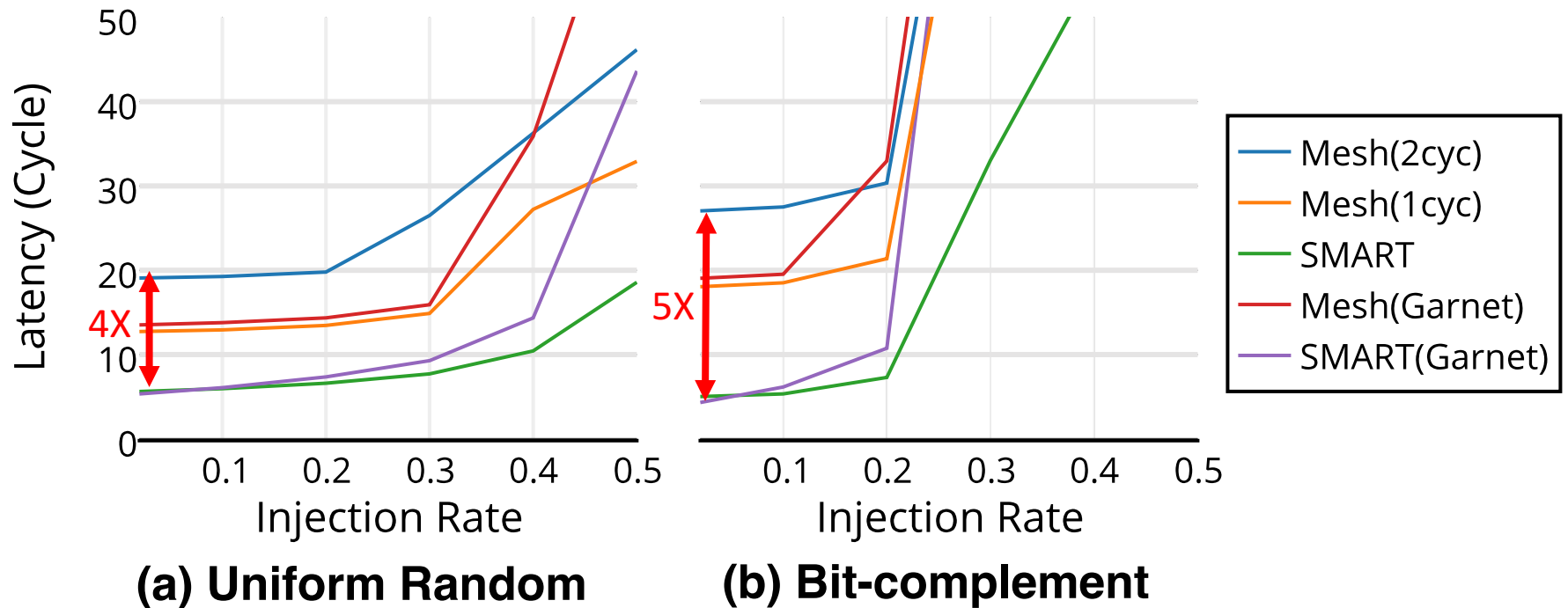
- **Routing Calculation**
  - **XY**, **YX**, and **source-routing**
  - **One-hot** encoding hop count + **shift-based** routing calculation
  - For SMART, routing calculation is done during bypasses
- **VC Selection**
  - **FIFO-based dynamic** VC selection
  - Next VC is stored in a separate register
  - For SMART, VC selection is done during bypasses

# Outline

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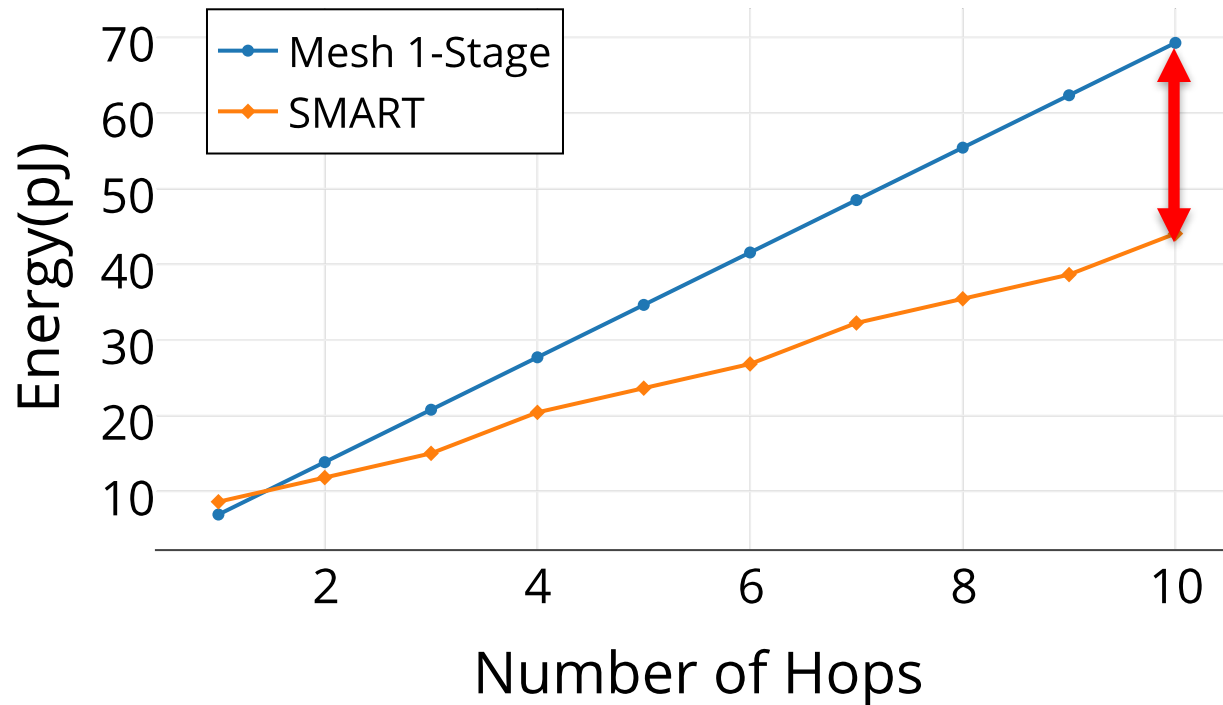
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# Latency



# Energy Consumption

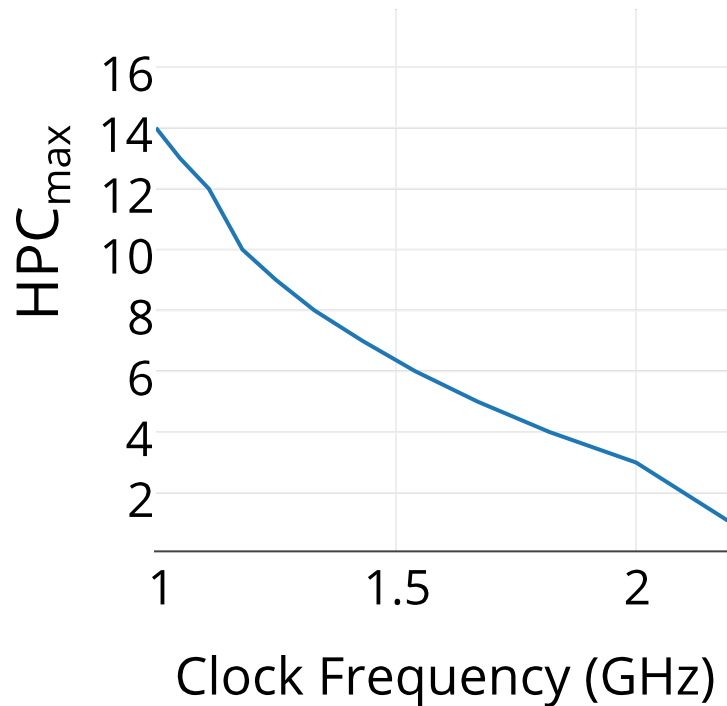
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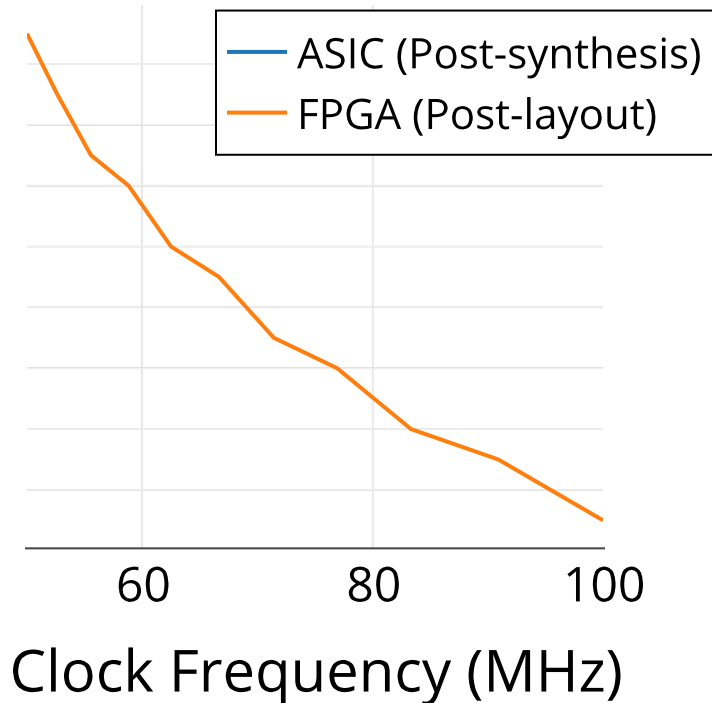
**Repeaters require less energy than clocked latches**

# HPCmax

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**(a) HPCmax on ASIC**



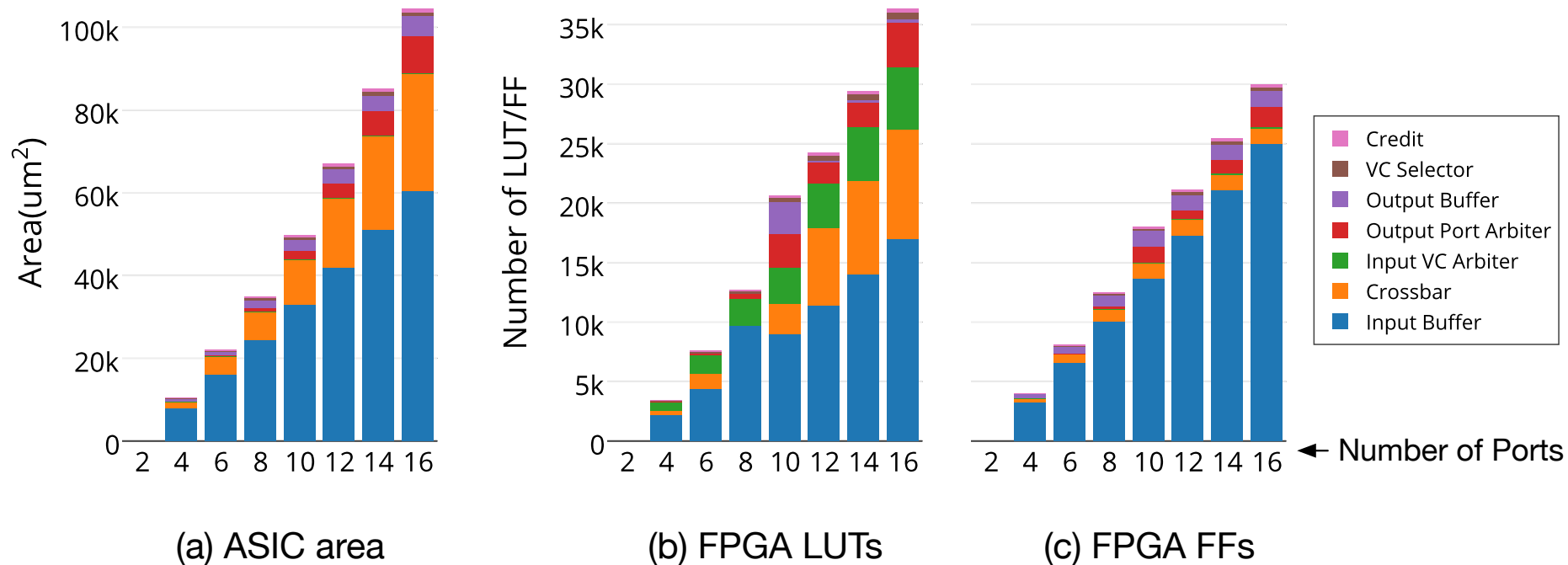
**(b) HPCmax on FPGA**

# Outline

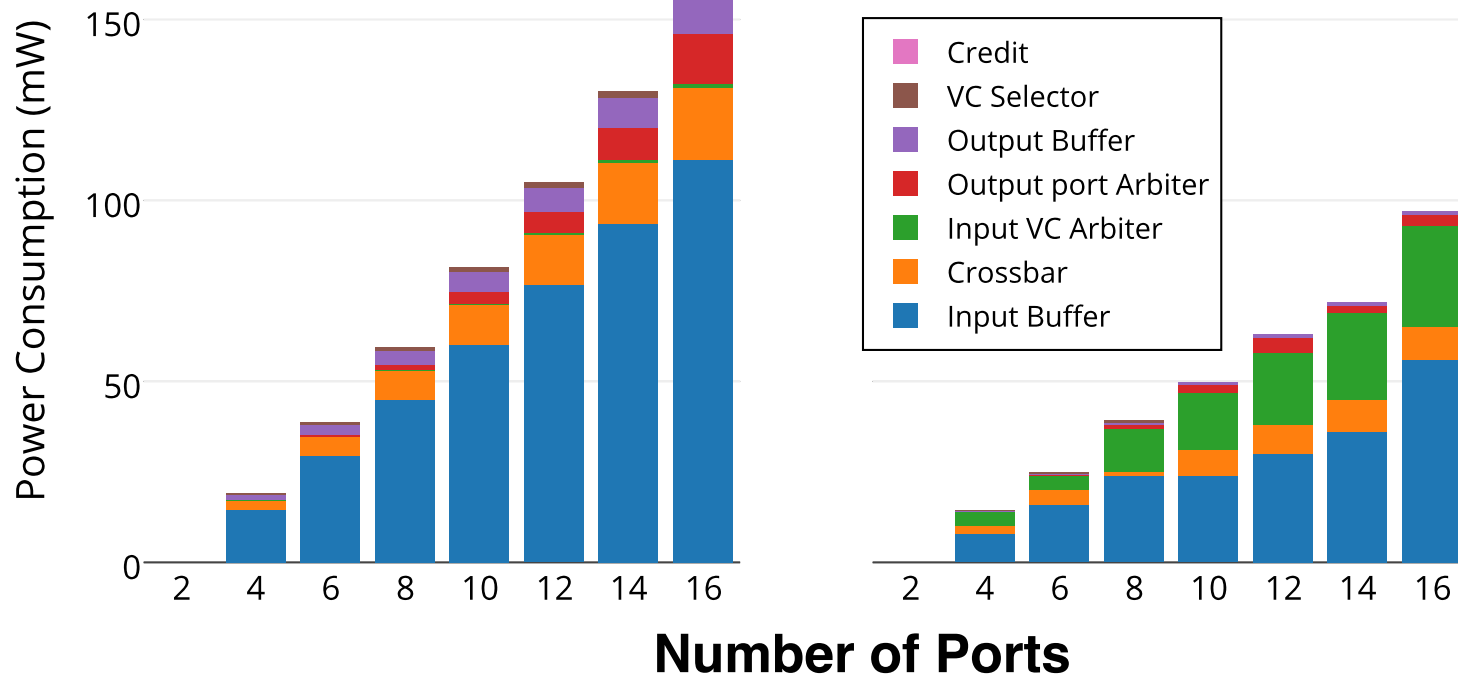
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# Router Area



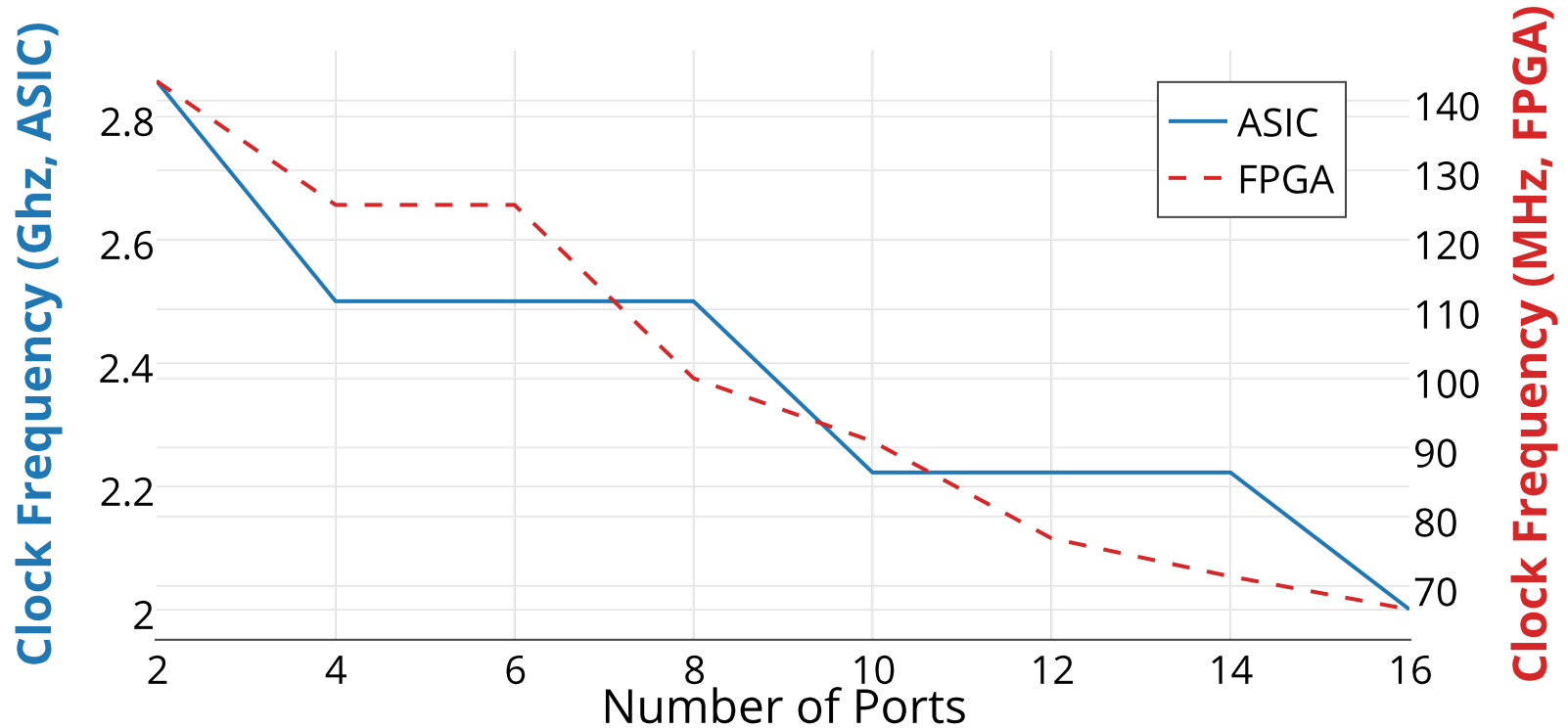
# Router Power



**(a) ASIC**

**(b) FPGA**

# Maximum Clock Frequency



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# Conclusion

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- NoCs are crucial components to support many-IP heterogeneous systems
  - Providing connectivity while satisfying their diverse requirements.
- ***OpenSMART*** provides automatic generation of NoCs for many-IP heterogeneous systems
  - Supports recent low latency SMART NoC as well as highly-optimized 1-cycle routers
  - Written in high-level HDLs

# Announcement

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- OpenSMART contributes the open-source hardware ecosystem!
- Source code will be available in May 2017
- Please sign up via our webpage to request the source code

<http://synergy.ece.gatech.edu/tools/opensmart/>

**Thank you!**